

DEPTH FIRST SEARCH (DFS)

1 Introduction

Depth First Search (DFS) is a fundamental graph traversal algorithm that explores a graph by moving as deep as possible along each branch before backtracking.

DFS can be implemented using:

- Recursive approach (implicit stack)
- Iterative approach (explicit stack)

This document explains **iterative DFS using a stack** with a detailed example and stack diagrams.

2 DFS Algorithm (Iterative Using Stack)

Algorithm 1 Depth First Search (Iterative)

Require: Graph $G(V, E)$, starting vertex s

Ensure: DFS traversal order

```
1: Create an empty stack  $S$ 
2: Mark all vertices as unvisited
3: Push  $s$  onto stack  $S$ 
4: while  $S$  is not empty do
5:    $v \leftarrow$  pop from  $S$ 
6:   if  $v$  is not visited then
7:     Mark  $v$  as visited
8:     Process  $v$ 
9:     for each adjacent vertex  $u$  of  $v$  (in reverse order) do
10:      if  $u$  is not visited then
11:        Push  $u$  onto  $S$ 
12:      end if
13:    end for
14:   end if
15: end while
```

3 Graph Example

Consider a graph with 8 vertices:

$$V = \{A, B, C, D, E, F, G, H\}$$

Adjacency list representation:

$A : B, C$

$B : D, E$

$C : F$

$D : H$

$E : -$

$F : G$

$G : -$

$H : -$

DFS traversal starts from vertex A .

4 DFS Traversal Order

The final DFS traversal order is:

$$A \rightarrow B \rightarrow D \rightarrow H \rightarrow E \rightarrow C \rightarrow F \rightarrow G$$

5 DFS Stack Iterations

Iteration 1: Push A

A

Top

Iteration 2: Pop A, Push C, B

C

B

Top

Iteration 3: Pop B, Push E, D

C

E

D

Top

Iteration 4: Pop D, Push H

C

E

H

Top

Iteration 5: Pop H

C

E

Top

Iteration 6: Pop E

C

Top

Iteration 7: Pop C, Push F

F

Top

Iteration 8: Pop F, Push G

G

Top

Iteration 9: Pop G (Stack Empty)

DFS traversal is complete.

6 Time and Space Complexity

- **Time Complexity:** $O(V + E)$
- **Space Complexity:** $O(V)$

7 Conclusion

Depth First Search efficiently explores graphs and is widely used in applications such as path finding, cycle detection, topological sorting, and connectivity analysis.